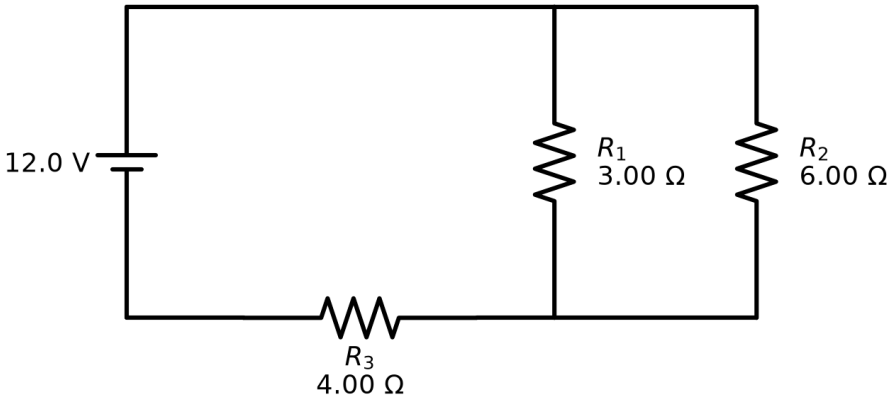
Series-parallel resistor networks: total current

Bank PHY2-DC-SPRTC-10222026 · created 10-29-2026 · 10 problems

Isomorphic numerical problems. Each item is a different series-parallel topology of a single battery and resistors; the student reduces the network to an equivalent resistance and applies Ohm's law to find the total current. The topology (and hence the reduction sequence) changes between items, not only the numbers.

Problem 1. The circuit shown is powered by a 12 V battery. Determine the total current supplied by the battery of the circuit shown.



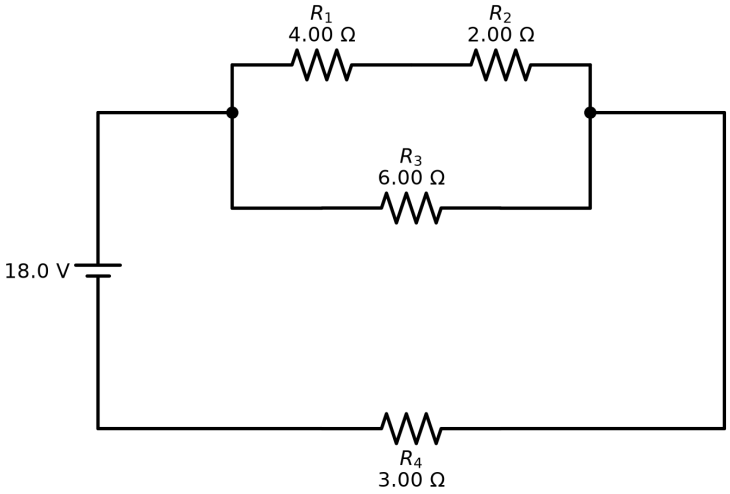
(A) 0.92 A

(B) 2.67 A

(C) 1.33 A

(D) 2.00 A

Problem 2. The circuit shown is powered by a 18 V battery. Determine the total current supplied by the battery of the circuit shown.



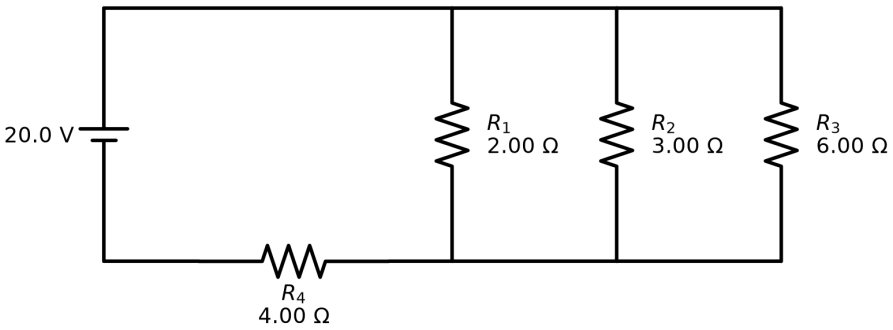
(A) 5.40 A

(B) 1.50 A

(C) 3.00 A

(D) 1.20 A

Problem 3. The circuit shown is powered by a 20 V battery. Determine the total current supplied by the battery of the circuit shown.



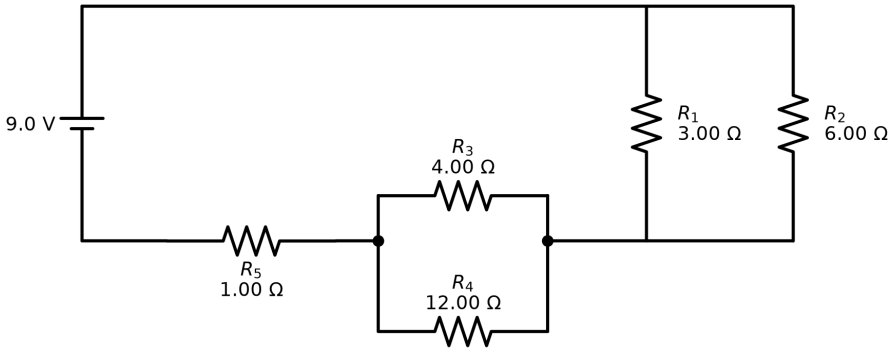
(A) 6.82 A

(B) 4.00 A

(C) 1.33 A

(D) 2.00 A

Problem 4. The circuit shown is powered by a 9 V battery. Determine the total current supplied by the battery of the circuit shown.



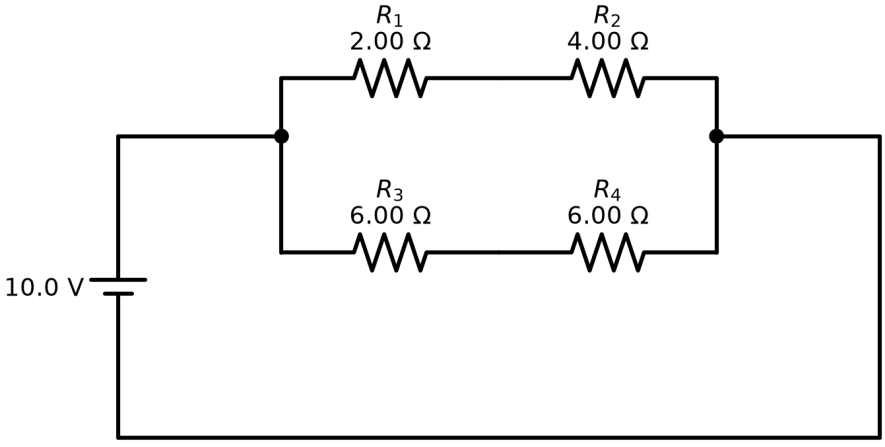
(A) 1.50 A

(B) 0.35 A

(C) 4.91 A

(D) 1.12 A

Problem 5. The circuit shown is powered by a 10 V battery. Determine the total current supplied by the battery of the circuit shown.



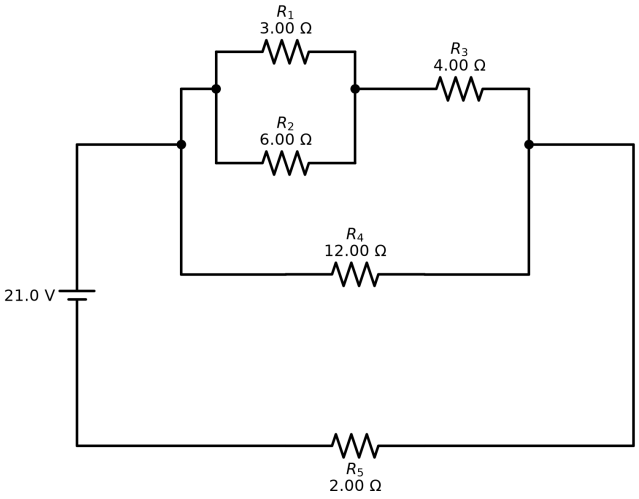
(A) 0.56 A

(B) 40.00 A

(C) 1.67 A

(D) 2.50 A

Problem 6. The circuit shown is powered by a 21 V battery. Determine the total current supplied by the battery of the circuit shown.



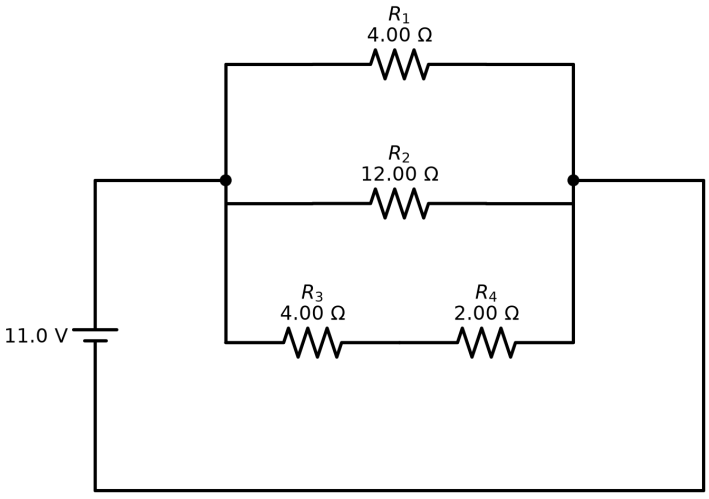
(A) 9.11 A

(B) 2.33 A

(C) 3.50 A

(D) 0.78 A

Problem 7. The circuit shown is powered by a 11 V battery. Determine the total current supplied by the battery of the circuit shown.



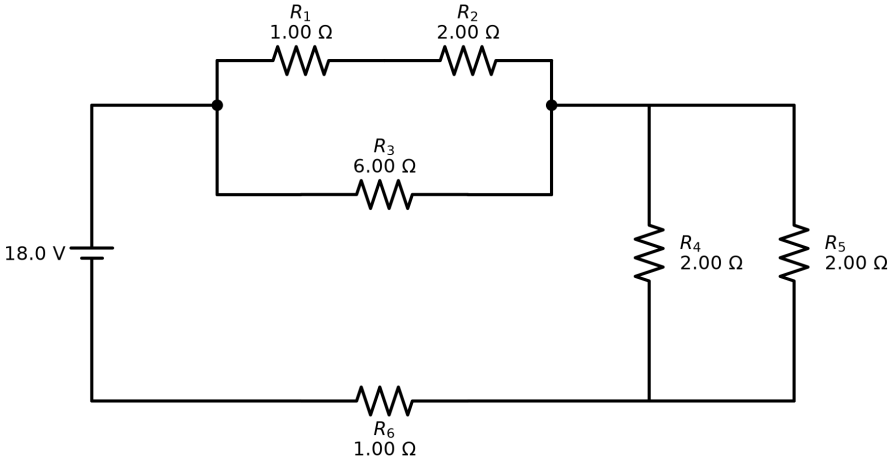
(A) 2.75 A

(B) 5.50 A

(C) 0.50 A

(D) 22.00 A

Problem 8. The circuit shown is powered by a 18 V battery. Determine the total current supplied by the battery of the circuit shown.



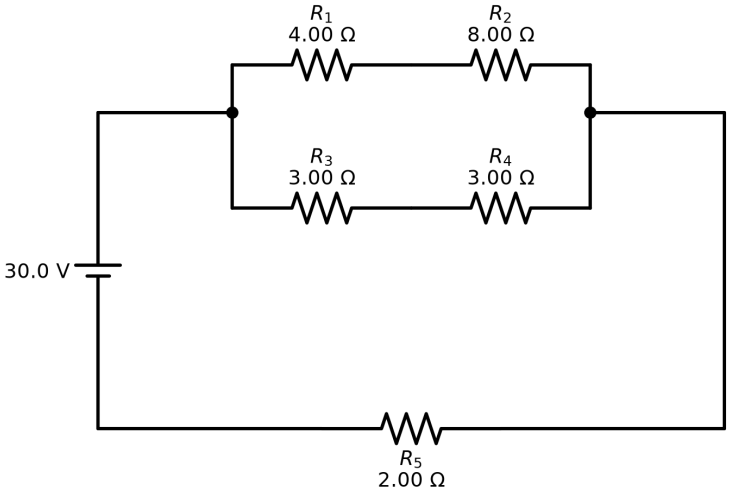
(A) 4.50 A

(B) 1.29 A

(C) 7.20 A

(D) 3.00 A

Problem 9. The circuit shown is powered by a 30 V battery. Determine the total current supplied by the battery of the circuit shown.



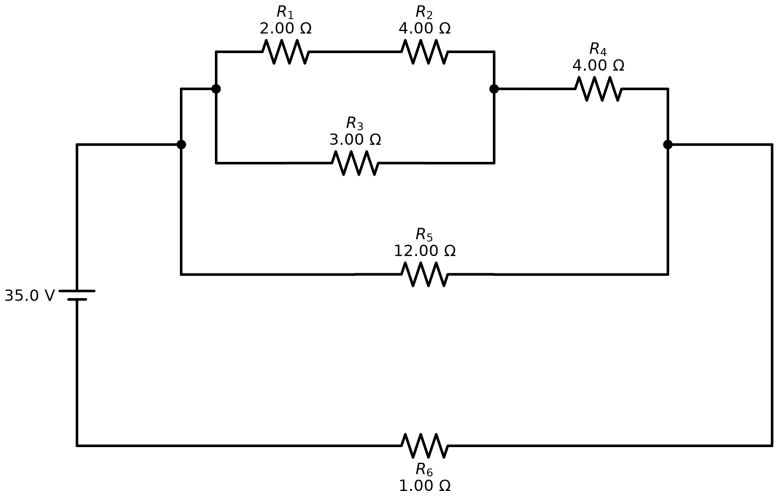
(A) 1.50 A

(B) 13.33 A

(C) 3.33 A

(D) 5.00 A

Problem 10. The circuit shown is powered by a 35 V battery. Determine the total current supplied by the battery of the circuit shown.



(A) 26.81 A

(B) 4.67 A

(C) 7.00 A

(D) 1.35 A

Answer key and worked solutions

Problem 1 — (D) 2.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in parallel: 1/R12 = 1/3.00 Ω + 1/6.00 Ω, so R12 = 2.00 Ω

2. R12 and R3 are in series: R123 = 2.00 Ω + 4.00 Ω = 6.00 Ω

Work back to the individual parts

1. The battery sees R123 = 6.00 Ω, so the total current is I = V / R = 12.00 V / 6.00 Ω = 2.00 A.

2. R12 and R3 are in series (R123), so each carries the same current 2.00 A; each voltage is I × R (0 V across a wire).

3. R1 and R2 are in parallel (R12), so each branch has the same voltage 4.00 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (3.00 Ω)	1.33 A	4.00 V	5.33 W
R2 (6.00 Ω)	0.67 A	4.00 V	2.67 W
R3 (4.00 Ω)	2.00 A	8.00 V	16.00 W
Battery	2.00 A	12.00 V	24.00 W (delivered)

Answer

- Total current from the battery = 2.00 A

Answer options (computed, each wrong option is one named mistake)

- 0.92 A: Student adds every resistor as if they were all in series.

- 2.67 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 1.33 A: Student reports the current through R1 (one branch) instead of the total.

- 2.00 A: Correct.

Problem 2 — (C) 3.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in series: R12 = 4.00 Ω + 2.00 Ω = 6.00 Ω

2. R12 and R3 are in parallel: 1/R123 = 1/6.00 Ω + 1/6.00 Ω, so R123 = 3.00 Ω

3. R123 and R4 are in series: R1234 = 3.00 Ω + 3.00 Ω = 6.00 Ω

Work back to the individual parts

1. The battery sees R1234 = 6.00 Ω, so the total current is I = V / R = 18.00 V / 6.00 Ω = 3.00 A.

2. R123 and R4 are in series (R1234), so each carries the same current 3.00 A; each voltage is I × R (0 V across a wire).

3. R12 and R3 are in parallel (R123), so each branch has the same voltage 9.00 V; each current is V / R (0 A through an open branch).

4. R1 and R2 are in series (R12), so each carries the same current 1.50 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (4.00 Ω)	1.50 A	6.00 V	9.00 W
R2 (2.00 Ω)	1.50 A	3.00 V	4.50 W
R3 (6.00 Ω)	1.50 A	9.00 V	13.50 W
R4 (3.00 Ω)	3.00 A	9.00 V	27.00 W
Battery	3.00 A	18.00 V	54.00 W (delivered)

Answer

- Total current from the battery = 3.00 A

Answer options (computed, each wrong option is one named mistake)

- 5.40 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 1.50 A: Student reports the current through R1 (one branch) instead of the total.

- 3.00 A: Correct.

- 1.20 A: Student adds every resistor as if they were all in series.

Problem 3 — (B) 4.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1, R2 and R3 are in parallel: 1/R123 = 1/2.00 Ω + 1/3.00 Ω + 1/6.00 Ω, so R123 = 1.00 Ω

2. R123 and R4 are in series: R1234 = 1.00 Ω + 4.00 Ω = 5.00 Ω

Work back to the individual parts

1. The battery sees R1234 = 5.00 Ω, so the total current is I = V / R = 20.00 V / 5.00 Ω = 4.00 A.

2. R123 and R4 are in series (R1234), so each carries the same current 4.00 A; each voltage is I × R (0 V across a wire).

3. R1, R2 and R3 are in parallel (R123), so each branch has the same voltage 4.00 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (2.00 Ω)	2.00 A	4.00 V	8.00 W
R2 (3.00 Ω)	1.33 A	4.00 V	5.33 W
R3 (6.00 Ω)	0.67 A	4.00 V	2.67 W
R4 (4.00 Ω)	4.00 A	16.00 V	64.00 W
Battery	4.00 A	20.00 V	80.00 W (delivered)

Answer

- Total current from the battery = 4.00 A

Answer options (computed, each wrong option is one named mistake)

- 6.82 A: Student treats series groups as parallel and parallel groups as series.

- 4.00 A: Correct.

- 1.33 A: Student adds every resistor as if they were all in series.

- 2.00 A: Student reports the current through R1 (one branch) instead of the total.

Problem 4 — (A) 1.50 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in parallel: 1/R12 = 1/3.00 Ω + 1/6.00 Ω, so R12 = 2.00 Ω

2. R3 and R4 are in parallel: 1/R34 = 1/4.00 Ω + 1/12.00 Ω, so R34 = 3.00 Ω

3. R12, R34 and R5 are in series: R12345 = 2.00 Ω + 3.00 Ω + 1.00 Ω = 6.00 Ω

Work back to the individual parts

1. The battery sees R12345 = 6.00 Ω, so the total current is I = V / R = 9.00 V / 6.00 Ω = 1.50 A.

2. R12, R34 and R5 are in series (R12345), so each carries the same current 1.50 A; each voltage is I × R (0 V across a wire).

3. R1 and R2 are in parallel (R12), so each branch has the same voltage 3.00 V; each current is V / R (0 A through an open branch).

4. R3 and R4 are in parallel (R34), so each branch has the same voltage 4.50 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (3.00 Ω)	1.00 A	3.00 V	3.00 W
R2 (6.00 Ω)	0.50 A	3.00 V	1.50 W
R3 (4.00 Ω)	1.12 A	4.50 V	5.06 W
R4 (12.00 Ω)	0.38 A	4.50 V	1.69 W
R5 (1.00 Ω)	1.50 A	1.50 V	2.25 W
Battery	1.50 A	9.00 V	13.50 W (delivered)

Answer

- Total current from the battery = 1.50 A

Answer options (computed, each wrong option is one named mistake)

- 1.50 A: Correct.

- 0.35 A: Student adds every resistor as if they were all in series.

- 4.91 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 1.12 A: Student reports the current through R3 (one branch) instead of the total.

Problem 5 — (D) 2.50 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in series: R12 = 2.00 Ω + 4.00 Ω = 6.00 Ω

2. R3 and R4 are in series: R34 = 6.00 Ω + 6.00 Ω = 12.00 Ω

3. R12 and R34 are in parallel: 1/R1234 = 1/6.00 Ω + 1/12.00 Ω, so R1234 = 4.00 Ω

Work back to the individual parts

1. The battery sees R1234 = 4.00 Ω, so the total current is I = V / R = 10.00 V / 4.00 Ω = 2.50 A.

2. R12 and R34 are in parallel (R1234), so each branch has the same voltage 10.00 V; each current is V / R (0 A through an open branch).

3. R1 and R2 are in series (R12), so each carries the same current 1.67 A; each voltage is I × R (0 V across a wire).

4. R3 and R4 are in series (R34), so each carries the same current 0.83 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (2.00 Ω)	1.67 A	3.33 V	5.56 W
R2 (4.00 Ω)	1.67 A	6.67 V	11.11 W
R3 (6.00 Ω)	0.83 A	5.00 V	4.17 W
R4 (6.00 Ω)	0.83 A	5.00 V	4.17 W
Battery	2.50 A	10.00 V	25.00 W (delivered)

Answer

- Total current from the battery = 2.50 A

Answer options (computed, each wrong option is one named mistake)

- 0.56 A: Student adds every resistor as if they were all in series.

- 40.00 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 1.67 A: Student reports the current through R1 (one branch) instead of the total.

- 2.50 A: Correct.

Problem 6 — (C) 3.50 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in parallel: 1/R12 = 1/3.00 Ω + 1/6.00 Ω, so R12 = 2.00 Ω

2. R12 and R3 are in series: R123 = 2.00 Ω + 4.00 Ω = 6.00 Ω

3. R123 and R4 are in parallel: 1/R1234 = 1/6.00 Ω + 1/12.00 Ω, so R1234 = 4.00 Ω

4. R1234 and R5 are in series: R12345 = 4.00 Ω + 2.00 Ω = 6.00 Ω

Work back to the individual parts

1. The battery sees R12345 = 6.00 Ω, so the total current is I = V / R = 21.00 V / 6.00 Ω = 3.50 A.

2. R1234 and R5 are in series (R12345), so each carries the same current 3.50 A; each voltage is I × R (0 V across a wire).

3. R123 and R4 are in parallel (R1234), so each branch has the same voltage 14.00 V; each current is V / R (0 A through an open branch).

4. R12 and R3 are in series (R123), so each carries the same current 2.33 A; each voltage is I × R (0 V across a wire).

5. R1 and R2 are in parallel (R12), so each branch has the same voltage 4.67 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (3.00 Ω)	1.56 A	4.67 V	7.26 W
R2 (6.00 Ω)	0.78 A	4.67 V	3.63 W
R3 (4.00 Ω)	2.33 A	9.33 V	21.78 W
R4 (12.00 Ω)	1.17 A	14.00 V	16.33 W
R5 (2.00 Ω)	3.50 A	7.00 V	24.50 W
Battery	3.50 A	21.00 V	73.50 W (delivered)

Answer

- Total current from the battery = 3.50 A

Answer options (computed, each wrong option is one named mistake)

- 9.11 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 2.33 A: Student reports the current through R3 (one branch) instead of the total.

- 3.50 A: Correct.

- 0.78 A: Student adds every resistor as if they were all in series.

Problem 7 — (B) 5.50 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R3 and R4 are in series: R34 = 4.00 Ω + 2.00 Ω = 6.00 Ω

2. R1, R2 and R34 are in parallel: 1/R1234 = 1/4.00 Ω + 1/12.00 Ω + 1/6.00 Ω, so R1234 = 2.00 Ω

Work back to the individual parts

1. The battery sees R1234 = 2.00 Ω, so the total current is I = V / R = 11.00 V / 2.00 Ω = 5.50 A.

2. R1, R2 and R34 are in parallel (R1234), so each branch has the same voltage 11.00 V; each current is V / R (0 A through an open branch).

3. R3 and R4 are in series (R34), so each carries the same current 1.83 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (4.00 Ω)	2.75 A	11.00 V	30.25 W
R2 (12.00 Ω)	0.92 A	11.00 V	10.08 W
R3 (4.00 Ω)	1.83 A	7.33 V	13.44 W
R4 (2.00 Ω)	1.83 A	3.67 V	6.72 W
Battery	5.50 A	11.00 V	60.50 W (delivered)

Answer

- Total current from the battery = 5.50 A

Answer options (computed, each wrong option is one named mistake)

- 2.75 A: Student reports the current through R1 (one branch) instead of the total.

- 5.50 A: Correct.

- 0.50 A: Student adds every resistor as if they were all in series.

- 22.00 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

Problem 8 — (A) 4.50 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in series: R12 = 1.00 Ω + 2.00 Ω = 3.00 Ω

2. R12 and R3 are in parallel: 1/R123 = 1/3.00 Ω + 1/6.00 Ω, so R123 = 2.00 Ω

3. R4 and R5 are in parallel: 1/R45 = 1/2.00 Ω + 1/2.00 Ω, so R45 = 1.00 Ω

4. R123, R45 and R6 are in series: R123456 = 2.00 Ω + 1.00 Ω + 1.00 Ω = 4.00 Ω

Work back to the individual parts

1. The battery sees R123456 = 4.00 Ω, so the total current is I = V / R = 18.00 V / 4.00 Ω = 4.50 A.

2. R123, R45 and R6 are in series (R123456), so each carries the same current 4.50 A; each voltage is I × R (0 V across a wire).

3. R12 and R3 are in parallel (R123), so each branch has the same voltage 9.00 V; each current is V / R (0 A through an open branch).

4. R1 and R2 are in series (R12), so each carries the same current 3.00 A; each voltage is I × R (0 V across a wire).

5. R4 and R5 are in parallel (R45), so each branch has the same voltage 4.50 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (1.00 Ω)	3.00 A	3.00 V	9.00 W
R2 (2.00 Ω)	3.00 A	6.00 V	18.00 W
R3 (6.00 Ω)	1.50 A	9.00 V	13.50 W
R4 (2.00 Ω)	2.25 A	4.50 V	10.12 W
R5 (2.00 Ω)	2.25 A	4.50 V	10.12 W
R6 (1.00 Ω)	4.50 A	4.50 V	20.25 W
Battery	4.50 A	18.00 V	81.00 W (delivered)

Answer

- Total current from the battery = 4.50 A

Answer options (computed, each wrong option is one named mistake)

- 4.50 A: Correct.

- 1.29 A: Student adds every resistor as if they were all in series.

- 7.20 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 3.00 A: Student reports the current through R1 (one branch) instead of the total.

Problem 9 — (D) 5.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in series: R12 = 4.00 Ω + 8.00 Ω = 12.00 Ω

2. R3 and R4 are in series: R34 = 3.00 Ω + 3.00 Ω = 6.00 Ω

3. R12 and R34 are in parallel: 1/R1234 = 1/12.00 Ω + 1/6.00 Ω, so R1234 = 4.00 Ω

4. R1234 and R5 are in series: R12345 = 4.00 Ω + 2.00 Ω = 6.00 Ω

Work back to the individual parts

1. The battery sees R12345 = 6.00 Ω, so the total current is I = V / R = 30.00 V / 6.00 Ω = 5.00 A.

2. R1234 and R5 are in series (R12345), so each carries the same current 5.00 A; each voltage is I × R (0 V across a wire).

3. R12 and R34 are in parallel (R1234), so each branch has the same voltage 20.00 V; each current is V / R (0 A through an open branch).

4. R1 and R2 are in series (R12), so each carries the same current 1.67 A; each voltage is I × R (0 V across a wire).

5. R3 and R4 are in series (R34), so each carries the same current 3.33 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (4.00 Ω)	1.67 A	6.67 V	11.11 W
R2 (8.00 Ω)	1.67 A	13.33 V	22.22 W
R3 (3.00 Ω)	3.33 A	10.00 V	33.33 W
R4 (3.00 Ω)	3.33 A	10.00 V	33.33 W
R5 (2.00 Ω)	5.00 A	10.00 V	50.00 W
Battery	5.00 A	30.00 V	150.00 W (delivered)

Answer

- Total current from the battery = 5.00 A

Answer options (computed, each wrong option is one named mistake)

- 1.50 A: Student adds every resistor as if they were all in series.

- 13.33 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 3.33 A: Student reports the current through R3 (one branch) instead of the total.

- 5.00 A: Correct.

Problem 10 — (C) 7.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in series: R12 = 2.00 Ω + 4.00 Ω = 6.00 Ω

2. R12 and R3 are in parallel: 1/R123 = 1/6.00 Ω + 1/3.00 Ω, so R123 = 2.00 Ω

3. R123 and R4 are in series: R1234 = 2.00 Ω + 4.00 Ω = 6.00 Ω

4. R1234 and R5 are in parallel: 1/R12345 = 1/6.00 Ω + 1/12.00 Ω, so R12345 = 4.00 Ω

5. R12345 and R6 are in series: R123456 = 4.00 Ω + 1.00 Ω = 5.00 Ω

Work back to the individual parts

1. The battery sees R123456 = 5.00 Ω, so the total current is I = V / R = 35.00 V / 5.00 Ω = 7.00 A.

2. R12345 and R6 are in series (R123456), so each carries the same current 7.00 A; each voltage is I × R (0 V across a wire).

3. R1234 and R5 are in parallel (R12345), so each branch has the same voltage 28.00 V; each current is V / R (0 A through an open branch).

4. R123 and R4 are in series (R1234), so each carries the same current 4.67 A; each voltage is I × R (0 V across a wire).

5. R12 and R3 are in parallel (R123), so each branch has the same voltage 9.33 V; each current is V / R (0 A through an open branch).

6. R1 and R2 are in series (R12), so each carries the same current 1.56 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (2.00 Ω)	1.56 A	3.11 V	4.84 W
R2 (4.00 Ω)	1.56 A	6.22 V	9.68 W
R3 (3.00 Ω)	3.11 A	9.33 V	29.04 W
R4 (4.00 Ω)	4.67 A	18.67 V	87.11 W
R5 (12.00 Ω)	2.33 A	28.00 V	65.33 W
R6 (1.00 Ω)	7.00 A	7.00 V	49.00 W
Battery	7.00 A	35.00 V	245.00 W (delivered)

Answer

- Total current from the battery = 7.00 A

Answer options (computed, each wrong option is one named mistake)

- 26.81 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 4.67 A: Student reports the current through R4 (one branch) instead of the total.

- 7.00 A: Correct.

- 1.35 A: Student adds every resistor as if they were all in series.